

海洋航行器六自由度非线性建模 (Fossen 1995 100% 全文无删减中英双语对照版)

海洋航行器六自由度非线性统一建模 (Nonlinear Modelling of Marine Vehicles in 6 Degrees of Freedom)

100% 全文无删减中英双语对照精读版 (Complete Unabridged Bilingual Edition)

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摘要 (Abstract)

[EN] In this paper a unified framework for vectorial parameterisation of inertia, Coriolis and centrifugal, and hydrodynamic added mass forces for marine vehicles in 6 degrees of freedom (DOF) is presented. Emphasis is placed on representing the 6 DOF nonlinear marine vehicle equations of motion in vector form satisfying certain matrix properties like symmetry, skew-symmetry and positive definiteness. For marine vehicles, this problem was first addressed by Fossen [1]. The proposed representations in this paper are based on extensions of this work. The main results are presented as two theorems. The first theorem is derived by applying Newton's equations whereas the second theorem reviews the main results of Sagatun and Fossen [6] where the Lagrangian formalism is applied.

[中文] 本文提出了一个用于海洋航行器六自由度 (6 DOF) 惯性力、科氏与向心力以及流体动力附加质量力矢量参数化的统一理论框架。本文重点在于将六自由度非线性海洋航行器运动方程表示为满足对称性、斜对称性（反对称性）以及正定性等特定代数矩阵性质的紧凑矢量形式。对于海洋航行器而言，该问题最初由 Fossen [1] 展开探索。本文提出的参数化表述是对其前期研究的进一步拓展与深化。核心研究成果体现为两大理论定理：第一定理通过应用牛顿力学运动方程严格推导得出；第二定理则系统回顾并归纳了 Sagatun 与 Fossen [6] 运用拉格朗日分析力学方法所取得的核心成果。

关键词 (Key words): 建模 (Modelling), 海洋航行器 (Marine vehicles), 运动方程 (Equations of motion), 拉格朗日力学与牛顿力学 (Lagrangian and Newtonian mechanics).

1. 引言 (1. Introduction)

[EN] In order to describe the motion of a marine vehicle in 6 degrees of freedom (DOF), 6 independent coordinates representing position and attitude are necessary. The 6 different motion components are defined as: surge, sway, heave, roll, pitch and yaw (see Table 1).

[中文] 为了精确描述海洋航行器在六自由度 (6 DOF) 空间中的全自由度运动，必须引入 6 个表示空间位置与姿态的独立广义坐标。这 6 个不同的运动分量标准定义为：纵荡 (surge)、横荡 (sway)、垂荡 (heave)、横滚 (roll)、俯仰 (pitch) 以及偏航 (yaw)（详见表 1）。

表 1：海洋航行器六自由度运动分量定义 (Table 1: 6 DOF motion components for a marine vehicle)

自由度 (DOF)	运动分量与物理方向 (Motions)	作用力与力矩 (Forces and Moments)	附体系速度 (Body-Fixed Velocities)	惯性系位置与欧拉角 (Inertial Position and Euler Angles)
1	motions in the x-direction (纵荡, surge)	X	u	x
2	motions in the y-direction (横荡, sway)	Y	v	y
3	motions in the z-direction (垂荡, heave)	Z	w	z
4	rotation about the x-axis (横滚, roll)	K	p	ϕ
5	rotation about the y-axis (俯仰, pitch)	M	q	θ
6	rotation about the z-axis (偏航, yaw)	N	r	ψ

[EN] Hence the general motion of a marine vehicle in 6 DOF can be described by the following vectors:

[中文] 因此，海洋航行器在六自由度空间中的一般运动状态可以用以下标准列矢量进行描述：

$$\begin{aligned} \boldsymbol{\eta} &= \begin{bmatrix} \eta_1 \\ \eta_2 \end{bmatrix}; \quad \boldsymbol{\eta}_1 = [x, y, z]^T; \quad \boldsymbol{\eta}_2 = [\phi, \theta, \psi]^T \\ \boldsymbol{\nu} &= \begin{bmatrix} \nu_1 \\ \nu_2 \end{bmatrix}; \quad \boldsymbol{\nu}_1 = [u, v, w]^T; \quad \boldsymbol{\nu}_2 = [p, q, r]^T \\ \boldsymbol{\tau} &= \begin{bmatrix} \tau_1 \\ \tau_2 \end{bmatrix}; \quad \boldsymbol{\tau}_1 = [X, Y, Z]^T; \quad \boldsymbol{\tau}_2 = [K, M, N]^T \end{aligned}$$

[EN] where $\boldsymbol{\eta}$ denotes the earth-fixed position and attitude vector, $\boldsymbol{\nu}$ denotes the body-fixed linear and angular velocity vector and $\boldsymbol{\tau}$ is used to describe the forces and moments acting on the vehicle in the body-fixed frame. The body-fixed and inertial reference frames are shown in Figure 1.

[中文] 其中 $\boldsymbol{\eta}$ 表示地球固连惯性坐标系（北-东-地 NED 坐标系）下的空间位置与欧拉角姿态矢量； $\boldsymbol{\nu}$ 表示固连于航行器机体的附体坐标系（前-右-下 FRD 坐标系）下的线速度与角速度矢量； $\boldsymbol{\tau}$ 用于描述在附体坐标系下作用于航行器机体上的广义力与力矩。附体参考坐标系 $\{b\}$ 与惯性参考坐标系 $\{n\}$ 的几何空间定义如图 1 所示。

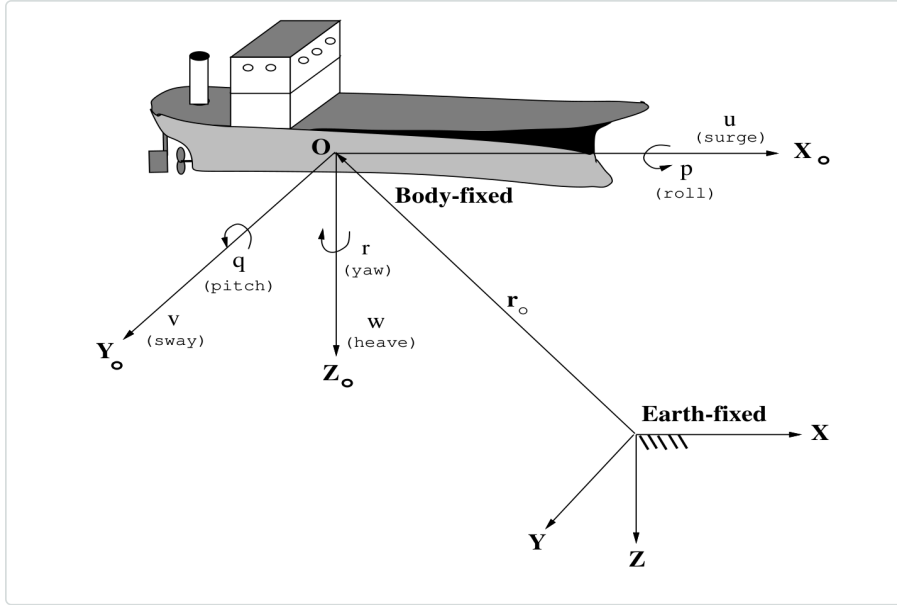


图 1 (Figure 1): 附体参考坐标系与地球固连惯性参考坐标系几何关系 (Body-fixed and earth-fixed reference frames)

[EN] The kinematic equations of motion relate the body-fixed velocities to the earth-fixed position and attitude rates through the transformations:

[中文] 运动学运动方程通过以下坐标变换矩阵将附体系速度与惯性系位置变化率及姿态角变化率联系起来:

$$\begin{aligned}\dot{\eta}_1 &= J_1(\eta_2)\nu_1, \quad \dot{\eta}_2 = J_2(\eta_2)\nu_2 \\ \dot{\eta} &= J(\eta)\nu = \begin{bmatrix} J_1(\eta_2) & \mathbf{0}_{3 \times 3} \\ \mathbf{0}_{3 \times 3} & J_2(\eta_2) \end{bmatrix} \begin{bmatrix} \nu_1 \\ \nu_2 \end{bmatrix}\end{aligned}$$

2. 牛顿力学方法 (2. Newtonian Approach)

[EN] The motion of a rigid body with respect to a body-fixed rotating reference frame $X_0Y_0Z_0$ with origin O is given by Newton's laws:

[中文] 刚体相对于原点为 O 的附体旋转参考坐标系 $X_0Y_0Z_0$ 的空间运动由牛顿运动定律严格给出:

$$m[\dot{\nu}_1 + \nu_2 \times \nu_1 + \dot{\nu}_2 \times \mathbf{r}_G + \nu_2 \times (\nu_2 \times \mathbf{r}_G)] = \boldsymbol{\tau}_1 \quad (1)$$

$$\mathbf{I}_0 \dot{\nu}_2 + \nu_2 \times (\mathbf{I}_0 \nu_2) + m \mathbf{r}_G \times (\dot{\nu}_1 + \nu_2 \times \nu_1) = \boldsymbol{\tau}_2 \quad (2)$$

[EN] where $\mathbf{r}_G = [x_G, y_G, z_G]^T$ is the centre of gravity and $\mathbf{I}_0$ is the inertia tensor with respect to the origin O of the body-fixed reference frame, that is:

[中文] 其中 $\mathbf{r}_G = [x_G, y_G, z_G]^T$ 为刚体质心 (重心) 在附体坐标系下的位置矢量, $\mathbf{I}_0$ 为刚体关于附体坐标系原点 O 的转动惯量张量, 即:

$$\mathbf{I}_0 = \begin{bmatrix} I_x & -I_{xy} & -I_{xz} \\ -I_{yx} & I_y & -I_{yz} \\ -I_{zx} & -I_{zy} & I_z \end{bmatrix}; \quad \mathbf{I}_0 = \mathbf{I}_0^T > 0; \quad \dot{\mathbf{I}}_0 = \mathbf{0} \quad (3)$$

[EN] Here I_x, I_y and I_z are the moments of inertia about the X_0, Y_0 and Z_0 axes and $I_{xy} = I_{yx}, I_{xz} = I_{zx}$ and $I_{yz} = I_{zy}$ are the products of inertia. These equations can be expressed in a more compact form as:

[中文] 这里 I_x, I_y 和 I_z 分别为绕附体坐标轴 X_0, Y_0 和 Z_0 的转动惯量, 而 $I_{xy} = I_{yx}, I_{xz} = I_{zx}$ 及 $I_{yz} = I_{zy}$ 为惯性积。上述六自由度运动方程可以表达为更为紧凑的矩阵向量形式:

$$\mathbf{M}_{RB}\dot{\boldsymbol{\nu}} + \mathbf{C}_{RB}(\boldsymbol{\nu})\boldsymbol{\nu} = \boldsymbol{\tau}_{RB} \quad (4)$$

[EN] where $\boldsymbol{\nu} = [u, v, w, p, q, r]^T$ is the body-fixed linear and angular velocity vector, $\boldsymbol{\tau}_{RB} = [X, Y, Z, K, M, N]^T$ is a generalised vector of external forces and moments and $\mathbf{M}_{RB}$ and $\mathbf{C}_{RB}(\boldsymbol{\nu})$ will be referred to as the rigid-body inertia and, Coriolis and centrifugal matrices, respectively. Before the parameterisation of these matrices are discussed it is convenient to define a skew-symmetric matrix operator.

[中文] 其中 $\boldsymbol{\nu} = [u, v, w, p, q, r]^T$ 为附体系线速度与角速度矢量; $\boldsymbol{\tau}_{RB} = [X, Y, Z, K, M, N]^T$ 为外部作用于刚体上的广义控制力与力矩矢量; $\mathbf{M}_{RB}$ 与 $\mathbf{C}_{RB}(\boldsymbol{\nu})$ 分别称为刚体惯性矩阵以及刚体科氏与向心力矩阵。在深入探讨这些矩阵的具体参数化方法之前, 引入反对称矩阵算子 (Skew-Symmetric Matrix Operator) 是非常必要且便利的。

定义 1 (反对称矩阵算子 / Definition 1: Skew-Symmetric Matrix Operator)

[EN] Let $S(\cdot)$ be a skew-symmetric matrix operator defined such that the vector cross product $\mathbf{a} \times \mathbf{b} = S(\mathbf{a})\mathbf{b}$, that is:

[中文] 设 $S(\cdot)$ 为反对称矩阵算子, 其定义满足向量叉乘运算 $\mathbf{a} \times \mathbf{b} = S(\mathbf{a})\mathbf{b}$, 即:

$$S(\mathbf{a}) = \begin{bmatrix} 0 & -a_3 & a_2 \\ a_3 & 0 & -a_1 \\ -a_2 & a_1 & 0 \end{bmatrix}; \quad S(\mathbf{a}) = -S^T(\mathbf{a}) \quad (5)$$

2.1 刚体惯性矩阵 (2.1 Rigid-Body Inertia Matrix)

[EN] The rigid-body inertia matrix $\mathbf{M}_{RB}$ can be uniquely determined from (1) and (2), that is:

[中文] 刚体惯性矩阵 $\mathbf{M}_{RB}$ 可以通过式 (1) 与式 (2) 唯一确定, 即满足加速度项的关系式:

$$\mathbf{M}_{RB}\dot{\boldsymbol{\nu}} = \begin{bmatrix} m\dot{\boldsymbol{\nu}}_1 + m\dot{\boldsymbol{\nu}}_2 \times \mathbf{r}_G \\ \mathbf{I}_0\dot{\boldsymbol{\nu}}_2 + m\mathbf{r}_G \times \dot{\boldsymbol{\nu}}_1 \end{bmatrix} \quad (6)$$

[EN] From this expression the positive definite inertia matrix $\mathbf{M}_{RB} = \mathbf{M}_{RB}^T > 0$ can be defined according to:

[中文] 由该表达式, 对称正定的刚体惯性矩阵 $\mathbf{M}_{RB} = \mathbf{M}_{RB}^T > 0$ 可规范定义为如下分块矩阵:

$$\mathbf{M}_{RB} = \begin{bmatrix} m\mathbf{I}_{3 \times 3} & -mS(\mathbf{r}_G) \\ mS(\mathbf{r}_G) & \mathbf{I}_0 \end{bmatrix}$$

[EN] Expanding this matrix yields the following 6×6 explicit representation:

[中文] 将该矩阵完全展开, 可得到如下 6×6 显式矩阵形式:

$$\mathbf{M}_{RB} = \begin{bmatrix} m & 0 & 0 & 0 & mz_G & -my_G \\ 0 & m & 0 & -mz_G & 0 & mx_G \\ 0 & 0 & m & my_G & -mx_G & 0 \\ 0 & -mz_G & my_G & I_x & -I_{xy} & -I_{xz} \\ mz_G & 0 & -mx_G & -I_{yx} & I_y & -I_{yz} \\ -my_G & mx_G & 0 & -I_{zx} & -I_{zy} & I_z \end{bmatrix} \quad (7)$$

[EN] where $\mathbf{I}_{3 \times 3}$ is the identity matrix and $\mathbf{I}_0$ is the inertia tensor with respect to O .

[中文] 其中 $\mathbf{I}_{3 \times 3}$ 为 3×3 单位矩阵, $\mathbf{I}_0$ 为刚体关于附体原点 O 的惯性张量。

2.2 刚体科氏与向心力矩阵 (2.2 Rigid-Body Coriolis and Centripetal Matrix)

[EN] The Coriolis and centripetal matrix can be represented in numerous ways. Two skew-symmetrical representations motivated from Newton's laws are given by Theorem 1.

[中文] 科氏与向心力矩阵可以有多种不同的参数化表示形式。由牛顿运动定律导出的两种斜对称（反对称）参数化表示形式由定理 1 给出。

定理 1 (科氏与向心力的参数化 / Theorem 1: Parameterisation of Coriolis and Centripetal Forces)

[EN] i) The Coriolis and centripetal matrix can be parameterised according to:

[中文] i) 形式一: 刚体科氏与向心力矩阵可参数化为如下分块矩阵形式:

$$\mathbf{C}_{RB}(\boldsymbol{\nu}) = \begin{bmatrix} \mathbf{0}_{3 \times 3} & -m\mathbf{S}(\boldsymbol{\nu}_1) - m\mathbf{S}(\boldsymbol{\nu}_2)\mathbf{S}(\mathbf{r}_G) \\ -m\mathbf{S}(\boldsymbol{\nu}_1) + m\mathbf{S}(\mathbf{r}_G)\mathbf{S}(\boldsymbol{\nu}_2) & -\mathbf{S}(\mathbf{I}_0\boldsymbol{\nu}_2) \end{bmatrix} \quad (8)$$

[EN] such that $\mathbf{C}_{RB}(\boldsymbol{\nu}) = -\mathbf{C}_{RB}^T(\boldsymbol{\nu})$ is skew-symmetrical. Notice that the matrix $\mathbf{S}(\boldsymbol{\nu}_2)\mathbf{S}(\mathbf{r}_G) = [\mathbf{S}(\mathbf{r}_G)\mathbf{S}(\boldsymbol{\nu}_2)]^T$. Also notice that $\mathbf{S}(\mathbf{0}) = \mathbf{0}$. Equation (8) can be written in component form according to:

[中文] 使得 $\mathbf{C}_{RB}(\boldsymbol{\nu}) = -\mathbf{C}_{RB}^T(\boldsymbol{\nu})$ 严格保持斜对称性。注意到矩阵关系 $\mathbf{S}(\boldsymbol{\nu}_2)\mathbf{S}(\mathbf{r}_G) = [\mathbf{S}(\mathbf{r}_G)\mathbf{S}(\boldsymbol{\nu}_2)]^T$, 同时显然满足 $\mathbf{S}(\mathbf{0}) = \mathbf{0}$ 。式 (8) 可以按各运动分量展开写成如下显式形式:

$$\mathbf{C}_{RB}(\boldsymbol{\nu}) = \begin{bmatrix} 0 & 0 & 0 & m(y_G q + z_G r) & -m(x_G q - w) & -m(x_G r + v) \\ 0 & 0 & 0 & -m(y_G p + w) & m(z_G r + x_G p) & -m(y_G r - u) \\ 0 & 0 & 0 & -m(z_G p - v) & -m(z_G q + u) & m(x_G p + y_G q) \\ -m(y_G q + z_G r) & m(y_G p + w) & m(z_G p - v) & 0 & -I_{yz}q - I_{xz}p + I_z r & I_{yz}r + I_{xy}p - I_y q \\ m(x_G q - w) & -m(z_G r + x_G p) & m(z_G q + u) & I_{yz}q + I_{xz}p - I_z r & 0 & -I_{xz}r - I_{xy}q + I_x p \\ m(x_G r + v) & m(y_G r - u) & -m(x_G p + y_G q) & -I_{yz}r - I_{xy}p + I_y q & I_{xz}r + I_{xy}q - I_x p & 0 \end{bmatrix} \quad (9)$$

[EN] ii) An alternative skew-symmetric representation is:

[中文] ii) 形式二: 另一种等价的斜对称分块参数化形式为:

$$\mathbf{C}_{RB}(\boldsymbol{\nu}) = \begin{bmatrix} \mathbf{0}_{3 \times 3} & -m\mathbf{S}(\boldsymbol{\nu}_1) - m\mathbf{S}(\boldsymbol{\nu}_2)\mathbf{r}_G \\ -m\mathbf{S}(\boldsymbol{\nu}_1) - m\mathbf{S}(\boldsymbol{\nu}_2)\mathbf{r}_G & m\mathbf{S}(\boldsymbol{\nu}_1)\mathbf{r}_G - \mathbf{S}(\mathbf{I}_0\boldsymbol{\nu}_2) \end{bmatrix} \quad (10)$$

[EN] which can be written in component form according to:

[中文] 该式可以表示为如下简洁分块形式:

$$\mathbf{C}_{RB}(\boldsymbol{\nu}) = \begin{bmatrix} \mathbf{0}_{3 \times 3} & \mathbf{C}_{12}(\boldsymbol{\nu}) \\ -\mathbf{C}_{12}^T(\boldsymbol{\nu}) & \mathbf{C}_{22}(\boldsymbol{\nu}) \end{bmatrix} \quad (11)$$

[EN] where:

[中文] 其中各 3×3 子矩阵分别为:

$$\mathbf{C}_{12}(\boldsymbol{\nu}) = \begin{bmatrix} 0 & m(w + qx_G - py_G) & -m(v - rx_G + pz_G) \\ -m(w - qx_G + py_G) & 0 & m(u + ry_G - qz_G) \\ m(v + rx_G - pz_G) & -m(u - ry_G + qz_G) & 0 \end{bmatrix} \quad (12)$$

$$\mathbf{C}_{22}(\boldsymbol{\nu}) = \begin{bmatrix} 0 & -I_{yz}q - I_{xz}p + I_z r + m(wx_G - uz_G) & I_{yz}r + I_{xy}p - I_y q - m(vx_G - uy_G) \\ I_{yz}q + I_{xz}p - I_z r - m(wx_G - uz_G) & 0 & -I_{xz}r - I_{xy}q + I_x p + m(wy_G - vz_G) \\ -I_{yz}r - I_{xy}p + I_y q + m(vx_G - uy_G) & I_{xz}r + I_{xy}q - I_x p - m(wy_G - vz_G) & 0 \end{bmatrix} \quad (13)$$

定理 1 证明过程 (Proof of Theorem 1)

[EN] **Proof:** The remaining terms in (1) and (2) after terms associated with $\dot{\boldsymbol{\nu}}_1$ and $\dot{\boldsymbol{\nu}}_2$ are removed can be written:

[中文] 证明: 从牛顿运动方程 (1) 和 (2) 中移去与加速度 $\dot{\boldsymbol{\nu}}_1$ 和 $\dot{\boldsymbol{\nu}}_2$ 相关的刚体惯性项后, 剩余的速度非线性项可表达为:

$$\mathbf{C}_{RB}(\boldsymbol{\nu})\boldsymbol{\nu} = \begin{bmatrix} m[\boldsymbol{\nu}_2 \times \boldsymbol{\nu}_1 + \boldsymbol{\nu}_2 \times (\boldsymbol{\nu}_2 \times \mathbf{r}_G)] \\ \boldsymbol{\nu}_2 \times (\mathbf{I}_0 \boldsymbol{\nu}_2) + m\mathbf{r}_G \times (\boldsymbol{\nu}_2 \times \boldsymbol{\nu}_1) \end{bmatrix} \quad (14)$$

[EN] By using the formulas:

[中文] 利用三维向量叉乘的下列恒等变换关系:

$$\mathbf{a} \times (\mathbf{b} \times \mathbf{c}) = S(\mathbf{a})S(\mathbf{b})\mathbf{c} \quad (15)$$

$$(\mathbf{a} \times \mathbf{b}) \times \mathbf{c} = S(S(\mathbf{a})\mathbf{b})\mathbf{c} \quad (16)$$

[EN] where $S(\mathbf{a})S(\mathbf{b}) \neq S(S(\mathbf{a})\mathbf{b})$, the first line in (14) can be rewritten to obtain the results under i) and ii) in Theorem 1 by noticing that:

[中文] 其中特别需要注意矩阵乘积 $S(\mathbf{a})S(\mathbf{b}) \neq S(S(\mathbf{a})\mathbf{b})$ 。通过注意以下向量恒等式, 式 (14) 的第一行可分别改写以证明定理 1 中的形式 i) 与形式 ii):

[EN] i) For case i):

[中文] i) 针对形式 i):

$$\begin{aligned} m[\boldsymbol{\nu}_2 \times \boldsymbol{\nu}_1 + \boldsymbol{\nu}_2 \times (\boldsymbol{\nu}_2 \times \mathbf{r}_G)] &= -m\boldsymbol{\nu}_1 \times \boldsymbol{\nu}_2 - m\boldsymbol{\nu}_2 \times (\mathbf{r}_G \times \boldsymbol{\nu}_2) \\ &= -mS(\boldsymbol{\nu}_1)\boldsymbol{\nu}_2 - mS(\boldsymbol{\nu}_2)S(\mathbf{r}_G)\boldsymbol{\nu}_2 \end{aligned} \quad (17)$$

[EN] ii) For case ii):

[中文] ii) 针对形式 ii):

$$\begin{aligned} m[\boldsymbol{\nu}_2 \times \boldsymbol{\nu}_1 + \boldsymbol{\nu}_2 \times (\boldsymbol{\nu}_2 \times \mathbf{r}_G)] &= -m\boldsymbol{\nu}_1 \times \boldsymbol{\nu}_2 - m(\boldsymbol{\nu}_2 \times \mathbf{r}_G) \times \boldsymbol{\nu}_2 \\ &= -mS(\boldsymbol{\nu}_1)\boldsymbol{\nu}_2 - mS(\boldsymbol{\nu}_2)\mathbf{r}_G \times \boldsymbol{\nu}_2 \end{aligned} \quad (18)$$

[EN] The second line in (14) can be rewritten in a similar manner. Moreover:

[中文] 式 (14) 的第二行力矩项可以按类似方法进行矢量展开。此外:

[EN] i) For case i):

[中文] i) 针对形式 i):

$$m\mathbf{r}_G \times (\boldsymbol{\nu}_2 \times \boldsymbol{\nu}_1) = mS(\mathbf{r}_G)S(\boldsymbol{\nu}_2)\boldsymbol{\nu}_1 \quad (19)$$

[EN] ii) For case ii), application of the Jacobi identity:

[中文] ii) 针对形式 ii)，应用向量分析中的雅可比恒等式 (Jacobi Identity)：

$$\mathbf{a} \times (\mathbf{b} \times \mathbf{c}) + \mathbf{b} \times (\mathbf{c} \times \mathbf{a}) + \mathbf{c} \times (\mathbf{a} \times \mathbf{b}) = \mathbf{0} \quad (20)$$

[EN] to the vector term $\mathbf{r}_G \times (\boldsymbol{\nu}_2 \times \boldsymbol{\nu}_1)$ yields:

[中文] 将其应用于力矩向量项 $\mathbf{r}_G \times (\boldsymbol{\nu}_2 \times \boldsymbol{\nu}_1)$ 可得：

$$\mathbf{r}_G \times (\boldsymbol{\nu}_2 \times \boldsymbol{\nu}_1) = -\boldsymbol{\nu}_1 \times (\mathbf{r}_G \times \boldsymbol{\nu}_2) - \boldsymbol{\nu}_2 \times (\boldsymbol{\nu}_1 \times \mathbf{r}_G) \quad (21)$$

[EN] Consequently:

[中文] 因此，该项可精确分解改写为：

$$\begin{aligned} m\mathbf{r}_G \times (\boldsymbol{\nu}_2 \times \boldsymbol{\nu}_1) &= -m(\boldsymbol{\nu}_2 \times \mathbf{r}_G) \times \boldsymbol{\nu}_1 + m(\boldsymbol{\nu}_1 \times \mathbf{r}_G) \times \boldsymbol{\nu}_2 \\ &= -mS(S(\boldsymbol{\nu}_2)\mathbf{r}_G)\boldsymbol{\nu}_1 + mS(S(\boldsymbol{\nu}_1)\mathbf{r}_G)\boldsymbol{\nu}_2 \end{aligned} \quad (22)$$

[EN] In addition to this $mS(\mathbf{0}) = \mathbf{0}$ and $\boldsymbol{\nu}_2 \times (\mathbf{I}_0\boldsymbol{\nu}_2) = -S(\mathbf{I}_0\boldsymbol{\nu}_2)\boldsymbol{\nu}_2$. This completes the proof of Theorem 1. ■

[中文] 此外，注意到 $mS(\mathbf{0}) = \mathbf{0}$ 且 $\boldsymbol{\nu}_2 \times (\mathbf{I}_0\boldsymbol{\nu}_2) = -S(\mathbf{I}_0\boldsymbol{\nu}_2)\boldsymbol{\nu}_2$ 。至此，定理 1 得证。■

定理 1 注记 (Remark to Theorem 1)

[EN] Two other skew-symmetric representations of $\mathbf{C}_{RB}(\boldsymbol{\nu})$ are obtained by omitting the zero term $S(\boldsymbol{\nu}_2)\boldsymbol{\nu}_2 = \mathbf{0}$ and by noticing that $S(\boldsymbol{\nu}_1)\boldsymbol{\nu}_2 = -S(\boldsymbol{\nu}_2)\boldsymbol{\nu}_1$. Hence, the results from Theorem 1 can be reformulated according to:

[中文] 通过省略恒零项 $S(\boldsymbol{\nu}_2)\boldsymbol{\nu}_2 = \mathbf{0}$ 并注意到反对称关系 $S(\boldsymbol{\nu}_1)\boldsymbol{\nu}_2 = -S(\boldsymbol{\nu}_2)\boldsymbol{\nu}_1$ ，还可以导出 $\mathbf{C}_{RB}(\boldsymbol{\nu})$ 的另外两种斜对称表达形式。据此，定理 1 的结果可重新整理如下：

[EN] i) First alternative representation:

[中文] i) 第一种备选形式：

$$\mathbf{C}_{RB}(\boldsymbol{\nu}) = \begin{bmatrix} mS(\boldsymbol{\nu}_2) & -mS(\boldsymbol{\nu}_2)S(\mathbf{r}_G) \\ mS(\mathbf{r}_G)S(\boldsymbol{\nu}_2) & -S(\mathbf{I}_0\boldsymbol{\nu}_2) \end{bmatrix} \quad (23)$$

[EN] ii) Second alternative representation:

[中文] ii) 第二种备选形式：

$$\mathbf{C}_{RB}(\boldsymbol{\nu}) = \begin{bmatrix} mS(\boldsymbol{\nu}_2) & -mS(S(\boldsymbol{\nu}_2)\mathbf{r}_G) \\ -mS(S(\boldsymbol{\nu}_2)\mathbf{r}_G) & mS(S(\boldsymbol{\nu}_1)\mathbf{r}_G) - S(\mathbf{I}_0\boldsymbol{\nu}_2) \end{bmatrix} \quad (24)$$

3. 拉格朗日力学方法 (3. Lagrangian Approach)

[EN] In order to include the effects due to hydrodynamic added mass it is advantageous to use a body-fixed energy formulation of Lagrange's equations since the vector $\int_0^t \boldsymbol{\nu} d\tau$ with $\boldsymbol{\nu} = [u, v, w, p, q, r]^T$ has no immediate physical interpretation. Moreover, Kirchhoff's equations which are derived from the ordinary Lagrange equations [3] will be applied for this purpose. Kirchhoff's equations in vector form are written [1]:

[中文] 为了将流体动力附加质量效应严密纳入动力学方程, 采用基于附体坐标系动能表述的拉格朗日力学方程具有显著优势, 因为积分向量 $\int_0^t \boldsymbol{\nu} d\tau$ (其中 $\boldsymbol{\nu} = [u, v, w, p, q, r]^T$) 并不具备直接的物理位移含义(准坐标性质)。为此, 本文将应用由普通拉格朗日方程推导而来的基尔霍夫方程(Kirchhoff's equations) [3]。基尔霍夫方程的紧凑矢量形式表述为 [1]:

$$\begin{aligned} \frac{d}{dt} \left(\frac{\partial T}{\partial \boldsymbol{\nu}_1} \right) + \boldsymbol{\nu}_2 \times \frac{\partial T}{\partial \boldsymbol{\nu}_1} &= \boldsymbol{\tau}_1 \\ \frac{d}{dt} \left(\frac{\partial T}{\partial \boldsymbol{\nu}_2} \right) + \boldsymbol{\nu}_2 \times \frac{\partial T}{\partial \boldsymbol{\nu}_2} + \boldsymbol{\nu}_1 \times \frac{\partial T}{\partial \boldsymbol{\nu}_1} &= \boldsymbol{\tau}_2 \end{aligned} \quad (25)$$

[EN] where T is the kinetic energy of the ambient water or the vehicle.

[中文] 其中 T 为周围流体介质或航行器刚体的系统动能。

定理 2 (由惯性矩阵确定科氏与向心力矩阵 / Theorem 2: Coriolis and Centripetal Matrix from Inertia Matrix)

[EN] Let $\boldsymbol{M}$ be an 6×6 inertia matrix defined as:

[中文] 设 $\boldsymbol{M}$ 为如下分块定义的 6×6 广义惯性矩阵:

$$\boldsymbol{M} = \begin{bmatrix} \boldsymbol{M}_{11} & \boldsymbol{M}_{12} \\ \boldsymbol{M}_{21} & \boldsymbol{M}_{22} \end{bmatrix} \quad (26)$$

[EN] Hence, the Coriolis and centripetal matrix can always be parameterised according to:

[中文] 据此, 科氏与向心力矩阵始终可以参数化为如下斜对称形式:

$$\boldsymbol{C}(\boldsymbol{\nu}) = \begin{bmatrix} \mathbf{0}_{3 \times 3} & -S(\boldsymbol{M}_{11}\boldsymbol{\nu}_1 + \boldsymbol{M}_{12}\boldsymbol{\nu}_2) \\ -S(\boldsymbol{M}_{11}\boldsymbol{\nu}_1 + \boldsymbol{M}_{12}\boldsymbol{\nu}_2) & -S(\boldsymbol{M}_{21}\boldsymbol{\nu}_1 + \boldsymbol{M}_{22}\boldsymbol{\nu}_2) \end{bmatrix} \quad (27)$$

[EN] where $\boldsymbol{C}(\boldsymbol{\nu}) = -\boldsymbol{C}^T(\boldsymbol{\nu})$. The results of Theorem 1 can be derived from this expression by letting $\boldsymbol{M} = \boldsymbol{M}_{RB}$.

[中文] 其中满足严格斜对称性 $\boldsymbol{C}(\boldsymbol{\nu}) = -\boldsymbol{C}^T(\boldsymbol{\nu})$ 。只要令广义惯性矩阵 $\boldsymbol{M} = \boldsymbol{M}_{RB}$, 定理 1 中的刚体科氏矩阵结果便可直接由此式推导导出。

定理 2 证明过程 (Proof of Theorem 2)

[EN] **Proof:** The proof follows Sagatun and Fossen [6] where the kinetic energy T is written as a quadratic form:

[中文] 证明: 本证明遵循 Sagatun 与 Fossen [6] 的推导思路, 其中系统动能 T 写为如下二次型形式:

$$T = \frac{1}{2} \boldsymbol{\nu}^T \mathbf{M} \boldsymbol{\nu} \quad (28)$$

[EN] Expanding this expression yields:

[中文] 将该二次型按分块矢量展开可得:

$$T = \frac{1}{2} (\boldsymbol{\nu}_1^T \mathbf{M}_{11} \boldsymbol{\nu}_1 + \boldsymbol{\nu}_1^T \mathbf{M}_{12} \boldsymbol{\nu}_2 + \boldsymbol{\nu}_2^T \mathbf{M}_{21} \boldsymbol{\nu}_1 + \boldsymbol{\nu}_2^T \mathbf{M}_{22} \boldsymbol{\nu}_2) \quad (29)$$

[EN] Hence, taking the partial derivatives with respect to the velocity vectors yields:

[中文] 据此, 对线速度和角速度矢量分别求偏导数 (广义动量), 可得:

$$\frac{\partial T}{\partial \boldsymbol{\nu}_1} = \mathbf{M}_{11} \boldsymbol{\nu}_1 + \mathbf{M}_{12} \boldsymbol{\nu}_2 \quad (30)$$

$$\frac{\partial T}{\partial \boldsymbol{\nu}_2} = \mathbf{M}_{21} \boldsymbol{\nu}_1 + \mathbf{M}_{22} \boldsymbol{\nu}_2 \quad (31)$$

[EN] From Kirchhoff's equations (25) it is recognised that:

[中文] 对比基尔霍夫方程 (25) 中的非线性速度项, 可以识别出:

$$\mathbf{C}(\boldsymbol{\nu}) \boldsymbol{\nu} = \begin{bmatrix} \boldsymbol{\nu}_2 \times \frac{\partial T}{\partial \boldsymbol{\nu}_1} \\ \boldsymbol{\nu}_2 \times \frac{\partial T}{\partial \boldsymbol{\nu}_2} + \boldsymbol{\nu}_1 \times \frac{\partial T}{\partial \boldsymbol{\nu}_1} \end{bmatrix} = \begin{bmatrix} \mathbf{0}_{3 \times 3} & -S \left(\frac{\partial T}{\partial \boldsymbol{\nu}_1} \right) \\ -S \left(\frac{\partial T}{\partial \boldsymbol{\nu}_1} \right) & -S \left(\frac{\partial T}{\partial \boldsymbol{\nu}_2} \right) \end{bmatrix} \begin{bmatrix} \boldsymbol{\nu}_1 \\ \boldsymbol{\nu}_2 \end{bmatrix} \quad (32)$$

[EN] which after substitution of (30) and (31) proves (27). ■

[中文] 将式 (30) 与式 (31) 代入式 (32) 中, 即严格证明了定理 2 的式 (27)。■

3.1 在周围流体与航行器系统中的应用 (3.1 Application to the Ambient Water and Vehicle Systems)

[EN] We will now use Theorem 2 to derive the rigid-body and ambient water Coriolis and centripetal matrices. In order to do this we will define an 6×6 added inertia matrix [7]:

[中文] 现在我们应用定理 2 来系统推导刚体以及周围流体介质的科氏与向心力矩阵。为此, 首先定义一个 6×6 的流体动力附加惯性 (附加质量) 矩阵 [7]:

$$\mathbf{M}_A = - \begin{bmatrix} X_{\dot{u}} & X_{\dot{v}} & X_{\dot{w}} & X_{\dot{p}} & X_{\dot{q}} & X_{\dot{r}} \\ Y_{\dot{u}} & Y_{\dot{v}} & Y_{\dot{w}} & Y_{\dot{p}} & Y_{\dot{q}} & Y_{\dot{r}} \\ Z_{\dot{u}} & Z_{\dot{v}} & Z_{\dot{w}} & Z_{\dot{p}} & Z_{\dot{q}} & Z_{\dot{r}} \\ K_{\dot{u}} & K_{\dot{v}} & K_{\dot{w}} & K_{\dot{p}} & K_{\dot{q}} & K_{\dot{r}} \\ M_{\dot{u}} & M_{\dot{v}} & M_{\dot{w}} & M_{\dot{p}} & M_{\dot{q}} & M_{\dot{r}} \\ N_{\dot{u}} & N_{\dot{v}} & N_{\dot{w}} & N_{\dot{p}} & N_{\dot{q}} & N_{\dot{r}} \end{bmatrix} \quad (33)$$

[EN] The notation of SNAME [7] is used in this expression; for instance the hydrodynamic added mass force Y_A along the y -axis due to an acceleration $\dot{u}$ in the x -direction is written as:

[中文] 该表达式采用了 SNAME [7] 标准流体动力学导数符号体系; 例如, 由于 x 轴方向加速度 $\dot{u}$ 在 y 轴方向诱导产生的流体动力附加质量力 Y_A 记为:

$$Y_A = Y_{\dot{u}}\dot{u} \quad \text{where} \quad Y_{\dot{u}} \triangleq \frac{\partial Y}{\partial \dot{u}} \quad (34)$$

[EN] For control applications we can assume that $\mathbf{M}_A > 0$ is constant and strictly positive. This is based on the assumption that $\mathbf{M}_A$ is independent of the wave frequency which is a good approximation for low-frequency control applications. For certain control applications like operation of underwater vehicles at great depth and positioning of surface ships we can also assume that $\mathbf{M}_A$ is symmetrical. However, this is not true for a surface ship moving at some speed [2]. For notational simplicity we will write:

[中文] 对于控制工程应用, 通常可以假设 $\mathbf{M}_A > 0$ 为常数且严格正定。这一假设基于附加质量独立于波浪频率的低频假设, 对于低频运动控制应用而言是极佳的物理近似。对于特定控制场景 (例如深水全潜作业的水下航行器 ROV/AUV, 以及处于动力定位工况的水面船舶), 我们还可以合理假设 $\mathbf{M}_A$ 是对称矩阵。然而, 对于以一定航速航行的水面船舶, 附加质量矩阵的对称性一般不再成立 [2]。为书写简便, 我们将矩阵记为分块形式:

$$\mathbf{M}_A = \begin{bmatrix} \mathbf{A}_{11} & \mathbf{A}_{12} \\ \mathbf{A}_{21} & \mathbf{A}_{22} \end{bmatrix}; \quad \mathbf{M}_{RB} = \begin{bmatrix} m\mathbf{I}_{3 \times 3} & -m\mathbf{S}(\mathbf{r}_G) \\ m\mathbf{S}(\mathbf{r}_G) & \mathbf{I}_0 \end{bmatrix} \quad (35)$$

[EN] for hydrodynamic added inertia and rigid-body inertia, respectively. Notice that the hydrodynamic added inertia and rigid-body inertia matrices are the same both in the Newtonian and Lagrangian approaches.

[中文] 分别表示流体动力附加惯性矩阵与刚体惯性矩阵。注意到无论是牛顿力学方法还是拉格朗日力学方法, 所导出的流体附加惯性矩阵与刚体惯性矩阵结构完全一致。

刚体科氏与向心力矩阵 (Rigid-Body Coriolis and Centripetal Matrix)

[EN] Theorem 2 with $\mathbf{M}_{RB}$ defined in (35) can now be used to derive the following expression for the rigid-body Coriolis and centripetal matrix:

[中文] 将式 (35) 中定义的 $\mathbf{M}_{RB}$ 代入定理 2, 即可导出如下刚体科氏与向心力矩阵表达式:

$$\mathbf{C}_{RB}(\boldsymbol{\nu}) = \begin{bmatrix} \mathbf{0}_{3 \times 3} & -m\mathbf{S}(\boldsymbol{\nu}_1) - m\mathbf{S}(\boldsymbol{\nu}_2)\mathbf{r}_G \\ -m\mathbf{S}(\boldsymbol{\nu}_1) - m\mathbf{S}(\boldsymbol{\nu}_2)\mathbf{r}_G & m\mathbf{S}(\boldsymbol{\nu}_1)\mathbf{r}_G - \mathbf{S}(\mathbf{I}_0\boldsymbol{\nu}_2) \end{bmatrix} \quad (36)$$

[EN] This is equivalent to case ii) of Theorem 1. Case i) can also be derived from this expression.

[中文] 该结果与定理 1 中的形式 ii) 完全等价。定理 1 中的形式 i) 亦可由此表达式进一步变形得出。

流体动力附加科氏与向心力矩阵 (Hydrodynamic Added Coriolis and Centripetal Matrix)

[EN] In addition to the rigid-body Coriolis and centripetal matrix, an expression for the hydrodynamic added Coriolis and centripetal matrix can be derived. Moreover, Theorem 2 with $\mathbf{M}_A$ defined in (35) yields:

[中文] 除了刚体科氏与向心力矩阵外，还可以精确推导出流体动力附加科氏与向心力矩阵。将式 (35) 中定义的 $\mathbf{M}_A$ 代入定理 2 可得：

$$\mathbf{C}_A(\boldsymbol{\nu}) = \begin{bmatrix} \mathbf{0}_{3 \times 3} & -S(\mathbf{A}_{11}\boldsymbol{\nu}_1 + \mathbf{A}_{12}\boldsymbol{\nu}_2) \\ -S(\mathbf{A}_{11}\boldsymbol{\nu}_1 + \mathbf{A}_{12}\boldsymbol{\nu}_2) & -S(\mathbf{A}_{21}\boldsymbol{\nu}_1 + \mathbf{A}_{22}\boldsymbol{\nu}_2) \end{bmatrix} \quad (37)$$

[EN] which can be written in component form according to:

[中文] 该式可以按各分量展开为如下显式形式：

$$\mathbf{C}_A(\boldsymbol{\nu}) = \begin{bmatrix} 0 & 0 & 0 & 0 & -a_3 & a_2 \\ 0 & 0 & 0 & a_3 & 0 & -a_1 \\ 0 & 0 & 0 & -a_2 & a_1 & 0 \\ 0 & -a_3 & a_2 & 0 & -b_3 & b_2 \\ a_3 & 0 & -a_1 & b_3 & 0 & -b_1 \\ -a_2 & a_1 & 0 & -b_2 & b_1 & 0 \end{bmatrix} \quad (38)$$

[EN] where:

[中文] 其中各系数分量定义为：

$$\begin{aligned} a_1 &= X_{\dot{u}}u + X_{\dot{v}}v + X_{\dot{w}}w + X_{\dot{p}}p + X_{\dot{q}}q + X_{\dot{r}}r \\ a_2 &= X_{\dot{v}}u + Y_{\dot{v}}v + Y_{\dot{w}}w + Y_{\dot{p}}p + Y_{\dot{q}}q + Y_{\dot{r}}r \\ a_3 &= X_{\dot{w}}u + Y_{\dot{w}}v + Z_{\dot{w}}w + Z_{\dot{p}}p + Z_{\dot{q}}q + Z_{\dot{r}}r \\ b_1 &= X_{\dot{p}}u + Y_{\dot{p}}v + Z_{\dot{p}}w + K_{\dot{p}}p + K_{\dot{q}}q + K_{\dot{r}}r \\ b_2 &= X_{\dot{q}}u + Y_{\dot{q}}v + Z_{\dot{q}}w + K_{\dot{q}}p + M_{\dot{q}}q + M_{\dot{r}}r \\ b_3 &= X_{\dot{r}}u + Y_{\dot{r}}v + Z_{\dot{r}}w + K_{\dot{r}}p + M_{\dot{r}}q + N_{\dot{r}}r \end{aligned} \quad (39)$$

[EN] An attractive simplification of this expression is to assume that:

[中文] 在工程实践中，一种极具吸引力且广泛应用的简化是假设附加质量矩阵为对角阵：

$$\mathbf{M}_A = -\text{diag}\{X_{\dot{u}}, Y_{\dot{v}}, Z_{\dot{w}}, K_{\dot{p}}, M_{\dot{q}}, N_{\dot{r}}\} \quad (40)$$

[EN] which simplifies $\mathbf{C}_A(\boldsymbol{\nu})$ to:

[中文] 此时，流体动力附加科氏与向心力矩阵可大幅简化为：

$$\mathbf{C}_A(\boldsymbol{\nu}) = \begin{bmatrix} 0 & 0 & 0 & 0 & -Z_{\dot{w}}w & Y_{\dot{v}}v \\ 0 & 0 & 0 & Z_{\dot{w}}w & 0 & -X_{\dot{u}}u \\ 0 & 0 & 0 & -Y_{\dot{v}}v & X_{\dot{u}}u & 0 \\ 0 & -Z_{\dot{w}}w & Y_{\dot{v}}v & 0 & -N_{\dot{r}}r & M_{\dot{q}}q \\ Z_{\dot{w}}w & 0 & -X_{\dot{u}}u & N_{\dot{r}}r & 0 & -K_{\dot{p}}p \\ -Y_{\dot{v}}v & X_{\dot{u}}u & 0 & -M_{\dot{q}}q & K_{\dot{p}}p & 0 \end{bmatrix} \quad (41)$$

3.2 总体运动方程 (3.2 Equations of Motion)

[EN] Assume that the external forces and moments τ_{RB} in (4) is written:

[中文] 假设式 (4) 中作用于刚体上的外部广义力和力矩 τ_{RB} 可以细分为如下物理力项之和:

$$\tau_{RB} = - \underbrace{M_A \dot{\nu} - C_A(\nu)\nu}_{\text{附加质量力项 (added mass)}} - \underbrace{D(\nu)\nu}_{\text{水动力阻尼力项 (damping)}} - \underbrace{g(\eta)}_{\text{恢复力项 (restoring forces)}} + \underbrace{\tau}_{\text{控制输入 (control input)}} \quad (42)$$

[EN] Hence, the marine vehicle equations of motion can be expressed according to:

[中文] 综合整理后, 海洋航行器在六自由度空间下的统一非线性运动方程可精确表达为:

$$M\dot{\nu} + C(\nu)\nu + D(\nu)\nu + g(\eta) = \tau \quad (43)$$

[EN] where:

[中文] 其中系统总惯性矩阵与总科氏矩阵分别为刚体项与流体附加项之和:

$$M = M_{RB} + M_A; \quad C(\nu) = C_{RB}(\nu) + C_A(\nu) \quad (44)$$

[EN] The matrix $D(\nu)$ can be interpreted as a non-symmetrical hydrodynamic damping matrix while $g(\eta)$ is included to describe buoyant and gravitational forces. Notice that these equations satisfy:

[中文] 矩阵 $D(\nu)$ 可解释为非对称的流体动力阻尼矩阵 (包含线性阻尼与非线性二次阻尼), 而矢量 $g(\eta)$ 用于描述浮力与重力共同作用产生的静水恢复力与力矩。极其重要的是, 上述系统矩阵满足以下严格的代数性质:

$$\dot{M} = 0; \quad M = M^T > 0; \quad C(\nu) = -C^T(\nu); \quad D(\nu) > 0 \quad (45)$$

[EN] For robot manipulators it is common to transform the desired state trajectory to joint space coordinates by applying the inverse kinematics. Hence the kinematic transformation matrix can be avoided in the control scheme. This is not possible for a marine vehicle since $\int_0^t \nu d\tau$ has no physical interpretation. Hence, the kinematic equations of motion must be included in the control design if position and attitude control are of interest. For this purpose it is common to apply an Euler angle representation (xyz-convention):

[中文] 对于工业机器人机械臂, 通常通过应用逆运动学将期望状态轨迹直接变换到关节空间坐标, 从而在控制律设计中可以完全规避运动学坐标变换矩阵。但对于海洋航行器而言这是不可行的, 因为附体系速度的时域积分 $\int_0^t \nu d\tau$ 不具备直接的几何物理含义 (属于不可积的非完整准坐标)。因此, 只要涉及空间位置与姿态的闭环控制, 就必须将运动学方程显式纳入控制系统设计中。为此, 工程上普遍采用欧拉角运动学表示 (遵循 xyz 转角约定):

$$\dot{\eta} = J(\eta)\nu \quad (46)$$

[EN] where $J(\eta) = \text{diag}\{J_1(\eta_2), J_2(\eta_2)\}$ and:

[中文] 其中 $J(\eta) = \text{diag}\{J_1(\eta_2), J_2(\eta_2)\}$, 其具体旋转与姿态变换矩阵为:

$$J_1(\eta_2) = \begin{bmatrix} c\psi c\theta & -s\psi c\theta + c\psi s\theta s\phi & s\psi s\theta + c\psi c\theta s\phi \\ s\psi c\theta & c\psi c\theta + s\psi s\theta s\phi & -c\psi s\theta + s\psi c\theta s\phi \\ -s\theta & c\theta s\phi & c\theta c\phi \end{bmatrix} \quad (47)$$

$$J_2(\eta_2) = \begin{bmatrix} 1 & s\phi t\theta & c\phi t\theta \\ 0 & c\phi & -s\phi \\ 0 & s\phi/c\theta & c\phi/c\theta \end{bmatrix} \quad (48)$$

[EN] Here $s \cdot = \sin(\cdot)$, $c \cdot = \cos(\cdot)$ and $t \cdot = \tan(\cdot)$. This representation is undefined for a pitch angle of $\theta = \pm 90^\circ$ (deg). The above system satisfies:

[中文] 此处简记 $s \cdot = \sin(\cdot)$, $c \cdot = \cos(\cdot)$ 及 $t \cdot = \tan(\cdot)$ 。需要注意，该欧拉角姿态变换矩阵在俯仰角 $\theta = \pm 90^\circ$ 处存在固有奇异点（分母为零无定义）。上述六自由度航行器动力学系统满足重要的能量无源性（Passivity）性质：

$$\boldsymbol{\nu}^T \mathbf{C}(\boldsymbol{\nu}) \boldsymbol{\nu} = 0; \quad \dot{\boldsymbol{\eta}}^T [\dot{\mathbf{M}}_{\boldsymbol{\eta}}(\boldsymbol{\eta}) - 2\mathbf{C}_{\boldsymbol{\eta}}(\boldsymbol{\eta}, \dot{\boldsymbol{\eta}})] \dot{\boldsymbol{\eta}} = 0 \quad (49)$$

[EN] where the earth-fixed inertia and Coriolis matrices are defined through the coordinate transformation:

[中文] 其中惯性系下的等效惯性矩阵与科氏矩阵通过运动学坐标变换定义为：

$$\mathbf{M}_{\boldsymbol{\eta}}(\boldsymbol{\eta}) = \mathbf{J}^{-T}(\boldsymbol{\eta}) \mathbf{M} \mathbf{J}^{-1}(\boldsymbol{\eta}); \quad \mathbf{C}_{\boldsymbol{\eta}}(\boldsymbol{\eta}, \dot{\boldsymbol{\eta}}) = \mathbf{J}^{-T}(\boldsymbol{\eta}) \left(\mathbf{C}(\boldsymbol{\nu}) - \mathbf{M} \mathbf{J}^{-1}(\boldsymbol{\eta}) \dot{\mathbf{J}}(\boldsymbol{\eta}) \right) \mathbf{J}^{-1}(\boldsymbol{\eta}) \quad (50)$$

[EN] which implies that the theory of passive adaptive control can be applied to design a stable control system.

[中文] 这一斜对称与能量守恒性质意味着，机器人学中成熟的基于无源性的自适应非线性控制理论（Passive Adaptive Control）可以直接推广应用于设计大范围全局渐近稳定的海洋航行器运动控制系统。

4. 结论 (4. Conclusions)

[EN] In this paper methods for parameterisation of nonlinear Coriolis, centripetal and inertia forces for marine vehicles in terms of matrix properties like symmetry, skew-symmetry and positiveness have been discussed. Emphases have been placed on deriving a model of the ambient water of the vehicle by considering the effects of hydrodynamic added inertia. The resulting model is written in vector form. Marine vehicle control design based on Lyapunov stability analyses for the proposed model structure is discussed in [1] and references therein.

[中文] 本文系统讨论了利用对称性、斜对称性（反对称性）以及正定性等优良矩阵代数性质对海洋航行器非线性科氏力、向心力以及惯性力进行参数化的理论方法。重点放在了通过深入考虑流体动力附加惯性效应来建立航行器周围流体介质的流固耦合动力学模型。所建立的统一模型全部采用严密的紧凑矢量形式表述。针对本文所提出的模型结构、基于李雅普诺夫稳定性理论的海洋航行器先进运动控制系统设计，已在文献 [1] 及其相关引用文献中展开了深入探讨。

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